

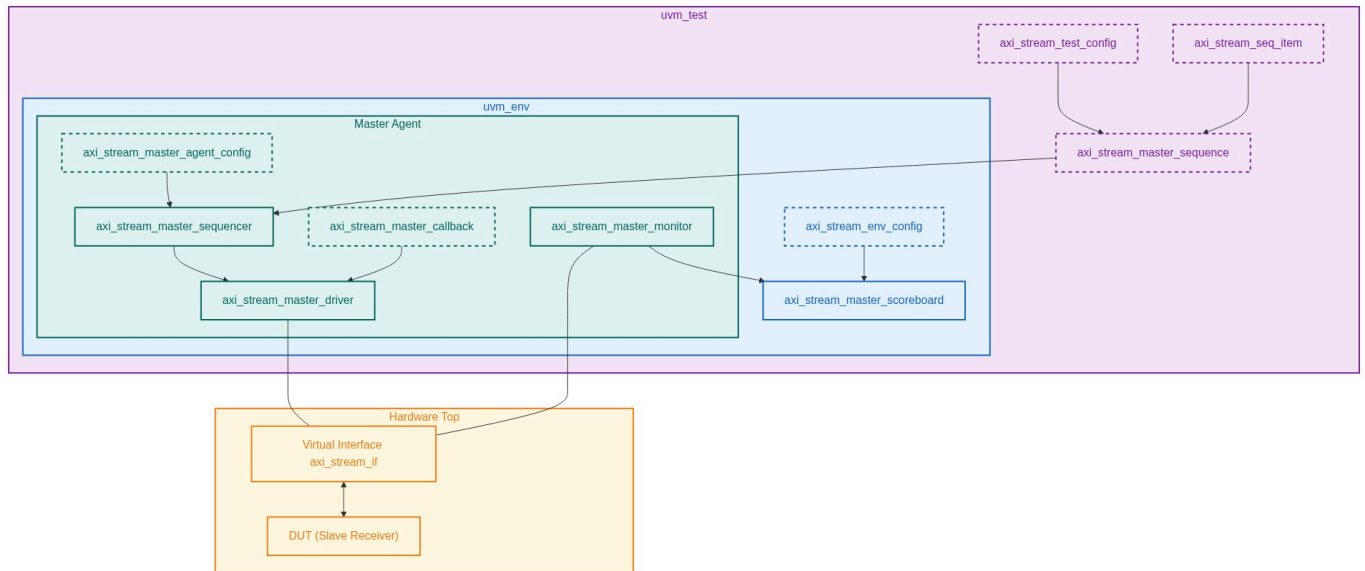
Verification Plan Report

Project: AMBA AXI-Stream Master Verification Plan

Version: 3.0

Verification Role: Master

1. UVM Environment Architecture



2. Test Requirements

Index	Feature/Sub	Name	Description
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REQ_MST_0 1	Handshake (TVALID Stability)	Transmitter Signal Stability During Stall	The AXI-Stream handshake protocol mandates strict signal stability from the Transmitter once TVALID is asserted. The payload (TDATA), sideband (TSTRB, TKEEP, TID, TDEST, TUSER), and control signal (TLAST) must all remain bit-for-bit identical across every ACLK cycle until the handshake occurs (TVALID AND TREADY both HIGH on the rising ACLK edge). Any deviation constitutes a protocol violation and results in an indeterminate transfer at the Receiver. Verification strategy targets the full combinatorial space of stall durations (1 to 512 cycles) using constrained randomization, while functional checking is performed via an SVA property (\$stable) monitoring all output signals. Negative scenarios inject deliberate bit-flips on TDATA/TUSER mid-stall to confirm the assertion fires correctly.
REQ_MST_0 2	Handshake (TVALID Independence)	Transmitter Must Not Gate TVALID on TREADY	The specification explicitly states that a Transmitter is NOT PERMITTED to wait until TREADY is asserted before asserting TVALID. This rule prevents deadlock conditions in interconnect topologies where a Receiver waits for TVALID before asserting TREADY, creating a circular dependency. The Master VIP driver must be configured to assert TVALID unconditionally as soon as a transaction is accepted from the sequencer. Verification must confirm that the time-delta between a sequence item being dispatched to the driver and TVALID being driven HIGH on the interface is zero or one clock cycle, and is never gated by the observed state of TREADY.

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REQ_MST_0 3	Reset (TVALID Suppression)	TVALID Must Be LOW During Active Reset	During assertion of ARESETn (LOW), the Master Transmitter must drive TVALID to a LOW state. All other signals may be driven to any value. This requirement ensures that no spurious transfers are initiated while the system is in an indeterminate state. Failure to comply can result in the Receiver latching garbage data on the first cycle after reset deassertion. Verification employs an SVA property that triggers on the falling edge of ARESETn and monitors TVALID for the entire duration of reset. Negative scenarios include asserting reset mid-transfer and verifying the Master VIP correctly suppresses TVALID within one clock cycle.
REQ_MST_0 4	Reset (Post-Reset Recovery)	TVALID Must Not Assert on Reset Deassertion Cycle	The specification requires that a Transmitter interface must ONLY begin driving TVALID HIGH at a rising ACLK edge FOLLOWING a rising edge at which ARESETn is observed HIGH. This mandatory one-cycle latency prevents spurious transfers that could occur if the Receiver's state machine has not yet completed its own reset recovery sequence. Verification targets the exact boundary condition: a directed test drives TVALID=1 on the same cycle as the ARESETn rising edge (a protocol violation) and confirms the SVA assertion fires. A second directed test confirms TVALID can be driven one cycle later without error.

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REQ_MST_0 5	Packet Integrity (Stream ID Consistency)	TID Must Remain Constant Within a Packet	The AXI-Stream specification defines a packet as a grouping of transfers sharing the same TID and TDEST values, terminated by TLAST=1. A Transmitter violating this rule (changing TID while TLAST is LOW) would cause downstream interconnect components to mis-route intermediate beats, resulting in data corruption at the destination. The merging of transfers with different TID values is explicitly forbidden. Verification uses a directed sequence that toggles TID mid-packet and confirms that the compliance monitor asserts an error. Functional tests verify that TID changes are only permitted on the beat immediately following a TLAST=1 assertion.
REQ_MST_0 6	Packet Integrity (Destination Consistency)	TDEST Must Remain Constant Within a Packet	Analogous to REQ_MST_05, the TDEST signal (routing information) must be held constant for all beats within a single packet. Changing TDEST before TLAST is asserted is a protocol violation that would cause the interconnect to attempt routing different beats of the same packet to different destinations, an illegal operation. Verification mirrors the TID consistency test suite, substituting TDEST as the target signal.
REQ_MST_0 7	Packet Integrity (Packet Boundary)	TLAST Marks the Final Beat of Each Packet	TLAST must be driven HIGH on the final beat of every packet. The count of TLAST assertions must be preserved between Transmitter and Receiver. Downstream components rely on this signal for buffer management, arbitration, and stream isolation. A deferred or missing TLAST can cause buffer overflow in interconnect FIFOs. Verification covers: single-beat packets (TLAST=1 on beat 1), multi-beat packets (TLAST on final beat only), back-to-back packets (no inter-packet gap required), and null packets (TLAST=1 with TKEEP all-zeros).

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REQ_MST_0 8	Byte Qualifiers (TKEEP Semantics)	TKEEP Must Correctly Flag Null Bytes	TKEEP identifies whether the content of an associated TDATA byte is processed as part of the data stream. A TKEEP bit of 0 indicates a null byte and must be preserved through interconnect. When TKEEP is absent, all bits default to HIGH, meaning all bytes are valid. Upsizing/downsizing interconnect components depend on TKEEP for correct packing operations. Verification sweeps all patterns: all-ones, all-zeros (null packet), walking-1, and randomly sparse configurations. Default value behavior (absent TKEEP defaults to all-HIGH) is tested via a configuration knob.
REQ_MST_0 9	Byte Qualifiers (TSTRB Semantics)	TSTRB Must Correctly Differentiate Data vs Position Bytes	TSTRB provides a byte-level qualifier that distinguishes data bytes (TSTRB=1) from position bytes (TSTRB=0). Position bytes carry positional information but not actual data payload. The default value rule specifies that if TSTRB is absent, TSTRB defaults to the value of TKEEP. Position-only transfers (all TSTRB=0) are a valid construct and must be correctly handled by the VIP model. Verification tests: full-data transfers (TSTRB=all 1), position-only transfers (TSTRB=0, TKEEP=1), mixed patterns, and the default-value rule (absent TSTRB equals TKEEP value).

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REQ_MST_1 0	Power Management (TWAKEUP Protocol (AXI5))	TWAKEUP Must Remain Asserted During Active Transfer	In AXI5-Stream mode, the TWAKEUP signal indicates activity on the interface and is intended for routing to a clock controller for power management. The specification mandates that if TWAKEUP and TVALID are both HIGH in the same cycle, TWAKEUP MUST remain asserted until TREADY is asserted (the handshake completes). Prematurely deasserting TWAKEUP could cause the clock controller to remove power or clocks before the transfer completes, creating an irreversible data loss scenario. Verification targets the concurrent assertion scenario, then injects an early TWAKEUP deassertion mid-stall and confirms the compliance assertion fires. Latency tests verify TWAKEUP can be asserted up to N cycles before TVALID (recommended: 1 cycle).
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3. Test Cases & Mapping

Index	Scenario	TR Mapping
TC_MST_001	Basic Handshake - Ready Before Valid: Drive TREADY=1 from Slave. Drive TVALID=1 from Master. Verify handshake completes on the same cycle. Monitor all output signals for 1 cycle.	REQ_MST_01, REQ_MST_02
TC_MST_002	Basic Handshake - Valid Before Ready: Drive TVALID=1. Hold TREADY=0 for 10 cycles. Assert TREADY=1. Verify handshake occurs on cycle 11 and TDATA was stable throughout all 10 stall cycles.	REQ_MST_01
TC_MST_003	Back-to-Back Single-Beat Transfers: Drive 100 consecutive single-beat packetss with no inter-packet gaps. TVALID=TREADY=1 for all cycles. Verify scoreboard receives 100 unique items.	REQ_MST_02
TC_MST_004	Randomized Stall Duration: Execute 200 transfers with constrained-random stall durations on TREADY (range: 0-50 cycles). Verify zero data discrepancies via scoreboard.	REQ_MST_01, REQ_MST_02
TC_MST_005	TDATA Stability Violation (Negative): Drive TVALID=1, TREADY=0. Inject a 1-bit flip on TDATA[0] on cycle 3 of the stall. Confirm REQ_MST_01 SVA assertion fires.	REQ_MST_01

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TC_MST_006	TUSER Stability Violation (Negative): Drive TVALID=1, TREADY=0. Inject a TUSER toggle on cycle 5 of the stall. Confirm stability assertion fires.	REQ_MST_01
TC_MST_007	Master Stall - Throttled TVALID: Interleave TVALID=0 and TVALID=1 every other cycle within a multi-beat packet. Verify packet integrity is maintained after each gap.	REQ_MST_02
TC_MST_008	Zero-Latency Handshake Verification: Assert TVALID and TREADY on the exact same ACLK cycle. Verify transfer completes in a single cycle with no pipeline delay in the scoreboard.	REQ_MST_01, REQ_MST_02
TC_MST_009	Initial Reset State: Apply ARESETn=0 at simulation start. Hold for 20 cycles. Verify TVALID=0 for all 20 cycles before any other stimulus is applied.	REQ_MST_03
TC_MST_010	Mid-Transfer Async Reset: Begin a multi-beat transfer. Assert ARESETn=0 mid-packet. Verify TVALID is suppressed within one clock cycle of reset assertion.	REQ_MST_03
TC_MST_011	Post-Reset Recovery - Legal: Deassert ARESETn. Wait exactly 2 cycles. Then drive TVALID=1. Verify no assertion fires and the transfer completes normally.	REQ_MST_04
TC_MST_012	Post-Reset Recovery - Illegal (Negative): Deassert ARESETn. Drive TVALID=1 on the very same ACLK cycle as the deassertion edge. Confirm the REQ_MST_04 SVA assertion fires.	REQ_MST_04
TC_MST_013	Repeated Reset Cycling: Cycle ARESETn 10 times with 5-cycle active periods. Verify the Master VIP returns to a clean IDLE state after each reset without TVALID glitches.	REQ_MST_03, REQ_MST_04
TC_MST_014	Single-Beat Packet: Drive a 1-beat transfer with TLAST=1. Verify the scoreboard correctly identifies it as a complete packet.	REQ_MST_07
TC_MST_015	Multi-Beat Packet (Fixed 16 beats): Drive 16 beats with TLAST=0 for beats 1-15 and TLAST=1 for beat 16. TID and TDEST constant throughout. Verify packet boundaries.	REQ_MST_05, REQ_MST_06, REQ_MST_07
TC_MST_016	Variable-Length Packets: Drive packets of constrained-random length (1 to 256 beats). Verify TLAST aligns with the last beat for each unique length across 50 iterations.	REQ_MST_07
TC_MST_017	Back-to-Back Packets (No Gap): Complete a 4-beat packet (TLAST=1 on beat 4). On the very next cycle, begin a new packet (TLAST=0). Verify no dead cycles between packets.	REQ_MST_07
TC_MST_018	Null Packet (Zero-Data): Drive TVALID=1, TLAST=1, TKEEP=all-zeros in one cycle. Verify the Slave DUT acknowledges the transfer and the scoreboard logs a zero-byte packet.	REQ_MST_07, REQ_MST_08

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TC_MST_019	TID Consistency Verification: Drive a 32-beat packet. Sample TID on every beat. Verify TID is bit-for-bit identical across all 32 beats. TID value is randomized per run.	REQ_MST_05
TC_MST_020	TID Violation (Negative): During beat 10 of a 20-beat packet (TLAST=0), change TID by toggling TID[0]. Confirm the REQ_MST_05 compliance assertion fires.	REQ_MST_05
TC_MST_021	TDEST Consistency Verification: Drive a 32-beat packet. Sample TDEST on every beat. Verify TDEST is constant. TDEST is randomized per run.	REQ_MST_06
TC_MST_022	TDEST Violation (Negative): During beat 5 of a 10-beat packet, toggle TDEST[0]. Confirm the REQ_MST_06 compliance assertion fires.	REQ_MST_06
TC_MST_023	Multi-Stream Interleaving: Send 3 back-to-back packets, each with a different TID (0x01, 0x02, 0x03). Verify the scoreboard correctly segregates and verifies each stream.	REQ_MST_05, REQ_MST_06
TC_MST_024	TKEEP All-Ones: Drive a standard 8-byte packet with TKEEP=0xFF. Verify all 8 bytes are processed by the scoreboard as valid data.	REQ_MST_08
TC_MST_025	TKEEP Walking-One Pattern: Drive 8 separate transfers, each with TKEEP using a walking-one pattern (0x01, 0x02, 0x04, 0x08, ...). Verify correct null-byte identification.	REQ_MST_08
TC_MST_026	Sparse TKEEP (Random Holes): Drive 100 packets with constrained-random TKEEP patterns. Verify scoreboard strips null bytes and reassembles data correctly.	REQ_MST_08
TC_MST_027	TSTRB Position-Only Transfer: Drive a beat where TSTRB=0x00 and TKEEP=0xFF. Verify the scoreboard identifies this as a position-only beat with no data bytes.	REQ_MST_09
TC_MST_028	TSTRB Mixed Data/Position: Drive a beat with alternating TSTRB bits (0xAA = 10101010). Verify the monitor correctly classifies odd-indexed bytes as data and even-indexed as position.	REQ_MST_09
TC_MST_029	TSTRB/TKEEP Default Rules: Configure the agent with TSTRB absent. Verify the monitor applies TSTRB=TKEEP default. Then verify the all-absent case defaults both to all-ones.	REQ_MST_08, REQ_MST_09
TC_MST_030	TWAKEUP Pre-Assertion: Assert TWAKEUP 1 cycle before TVALID. Verify TWAKEUP remains HIGH through the entire transfer and is deasserted after TREADY goes HIGH.	REQ_MST_10
TC_MST_031	TWAKEUP Concurrent Assertion: Assert TWAKEUP and TVALID on the same ACLK cycle. Hold TREADY=0 for 5 cycles. Verify TWAKEUP remains asserted for all 5 stall cycles.	REQ_MST_10

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TC_MST_032	TWAKEUP Early Drop Violation (Negative): Assert TWAKEUP and TVALID on the same cycle. Drop TWAKEUP on cycle 3 while TREADY remains LOW. Confirm the REQ_MST_10 assertion fires.	REQ_MST_10
TC_MST_033	TWAKEUP Without TVALID: Assert TWAKEUP for 10 cycles before TVALID. Verify interface is dormant (TVALID=0). Verify no transfers occur during this power-up window.	REQ_MST_10
TC_MST_034	Full-Stress Regression: Execute 10,000 randomized transfers with all signals constrained-random (TKEEP, TSTRB, TID, TDEST, TUSER, packet length, TREADY stalls). Scoreboard must achieve zero mismatches.	REQ_MST_01, REQ_MST_05, REQ_MST_07, REQ_MST_08
TC_MST_035	Max Data Width Throughput: Configure TDATA_WIDTH=1024. Run 500 back-to-back max-width transfers. Verify performance (throughput >= 99% utilization) and zero scoreboard errors.	REQ_MST_01, REQ_MST_02

4. Functional Coverage Groups

Group Name	Description
cg_handshake_states	Exhaustive cross-product of {TVALID=0/1} x {TREADY=0/1}. All 4 states must be hit.
cg_stall_duration	TREADY stall duration bins: [0, 1, 2-5, 6-20, 21-100, 101+] cycles.
cg_reset_exit_timing	Time delta between ARESETn rising edge and first TVALID assertion. Bins: [illegal: 0 cycles], [legal: 1 cycle], [relaxed: 2+ cycles].
cg_packet_length	Packet beat-count bins: [1], [2-4], [5-16], [17-64], [65-256], [257+].
cg_tid_tdest_cross	Cross-product of TID[3:0] x TDEST[3:0] combinations across packet boundaries.
cg_tkeep_patterns	TKEEP coverage: all-ones, all-zeros, walking-1, walking-0, random-sparse.
cg_tstrb_patterns	TSTRB coverage: all-data (1s), all-position (0s), alternating (0xAA, 0x55).
cg_twakeup_timing	TWAKEUP assertion timing relative to TVALID: [-3, -2, -1, 0, +1] cycles offset.

5. Interface Signals

Signal Name	Direction	Width	Description
ACLK	Input	1	Global clock. All signals are sampled on the rising edge of ACLK.
ARESETn	Input	1	Active-LOW global reset. Asserted asynchronously, deasserted synchronously to ACLK.
TVALID	Output	1	Driven by Master. Indicates that TDATA and all sideband signals are valid and stable.
TREADY	Input	1	Driven by Slave DUT. Indicates that the Receiver is ready to accept a transfer.

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TDATA	Output	TDATA_WIDTH	Primary payload data. Width must be an integer number of bytes (8,16,32,64,...,1024-bit).
TSTRB	Output	TDATA_WIDTH /8	Byte qualifier. Identifies whether content of associated TDATA byte is a data or position byte.
TKEEP	Output	TDATA_WIDTH /8	Byte qualifier. Identifies whether associated TDATA byte forms part of the data stream.
TLAST	Output	1	Packet boundary marker. When HIGH, signals the final beat of a packet.
TID	Output	TID_WIDTH	Data stream identifier. TID_WIDTH recommended to be no more than 8 bits.
TDEST	Output	TDEST_WIDTH	Routing information for the data stream. TDEST_WIDTH recommended to be no more than 8 bits.
TUSER	Output	TUSER_WIDTH	User-defined sideband. TUSER_WIDTH recommended as integer multiple of TDATA_WIDTH/8.
TWAKEUP	Output	1	AXI5-only. Indicates activity on the AXI-Stream interface for power management.

6. Verification Environment Structure

axi_stream_master_vip_tb/

```

|-- master_agent/
|   |-- axi_stream_master_driver.sv
|   |-- axi_stream_master_monitor.sv
|   |-- axi_stream_master_sequencer.sv
|   |-- axi_stream_master_agent.sv
|   |-- axi_stream_master_agent_config.sv
|   |-- axi_stream_master_callback.sv
|-- env/
|   |-- axi_stream_master_env.sv
|   |-- axi_stream_master_scoreboard.sv
|   |-- axi_stream_master_env_config.sv
|-- sequences/
|   |-- axi_stream_master_seq_item.sv
|   |-- axi_stream_master_base_sequence.sv
|   |-- axi_stream_master_rand_sequence.sv
|   |-- axi_stream_master_stress_sequence.sv
|-- top/
|   |-- axi_stream_master_tb_top.sv
|   |-- axi_stream_master_test.sv
|   |-- axi_stream_master_defines.sv
|   |-- axi_stream_if.sv

```

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7. SystemVerilog Implementation

axi_stream_master_seq_item.sv (Architecture Blueprint)

```
COMPONENT: axi_stream_master_seq_item
EXTENDS: uvm_sequence_item
MEMBERS:
  - [rand logic [TDATA_WIDTH-1:0]] tdata
  - [rand logic [TDATA_WIDTH/8-1:0]] tkeep
  - [rand logic [TDATA_WIDTH/8-1:0]] tstrb
  - [rand logic] tlast
  - [rand logic [TID_WIDTH-1:0]] tid
  - [rand logic [TDEST_WIDTH-1:0]] tdest
  - [rand logic [TUSER_WIDTH-1:0]] tuser
  - [rand int unsigned] stall_cycles
CONSTRAINTS:
  - stall_cycles: between 0 and 50 cycles (weighted toward 0 for high-throughput scenarios)
  - tkeep: cannot be all-zero unless tlast is also 1 (null packet rule)
  - tstrb: must be a subset of tkeep (no strobe bit set where keep is 0)
```

axi_stream_master_driver.sv (Architecture Blueprint)

```
COMPONENT: axi_stream_master_driver
EXTENDS: uvm_driver (axi_stream_master_seq_item)
CONFIG_REF: axi_stream_master_agent_config (Source: ConfigDB)
MEMBERS:
  - [virtual axi_stream_if] vif
  - [axi_stream_master_agent_config] cfg
METHODS & LOGIC:
  - build_phase: Retrieve vif from ConfigDB. Retrieve cfg from ConfigDB.
  - run_phase: Loop forever. Call seq_item_port.get_next_item(req).
    Drive vif.TVALID=1 and all payload pins from req on next posedge ACLK.
    Wait for TREADY=1 on the same posedge (handshake check).
    Call seq_item_port.item_done().
    If req.stall_cycles > 0, drive TVALID=0 for that many cycles before next item.
```

axi_stream_master_monitor.sv (Architecture Blueprint)

```
COMPONENT: axi_stream_master_monitor
EXTENDS: uvm_monitor
MEMBERS:
  - [virtual axi_stream_if] vif
  - [uvm_analysis_port] ap
METHODS & LOGIC:
  - run_phase: Loop on posedge ACLK. If ARESETn is LOW, confirm TVALID=0 else fire error.
    If TVALID=1 and TREADY=1 (handshake): capture all signals into new seq_item. Write to ap.
    If TVALID=1 and TREADY=0 (stall): verify $stable for all output signals every cycle.
```